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Analysis and Improvement of an Assembly Line in the Automotive Industry

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Abstract

In a market as competitive as the automotive industry, it becomes increasingly important for the organizations to adopt a culture of continuous improvement, which should cross-over all stakeholders in the organization. The continuous improvement of the processes, the increase in efficiency, and the elimination of waste, leads to a considerable increase in market competitiveness, not only economically, but also technologically. The focus of this work was the optimization of a production line, with the main goal being the increase of its productive capacity, so that it can comply with customer's requests. Thus, it was defined as a goal of this project: to increase the productive capacity to 1800 parts/day. The methodologies used were based on several continuous improvements and lean techniques, such as line balancing, standard work, visual management and 5S. The work developed allowed an increase of 37% of the production line capacity and an increase of 22% in the OEE of the production line.

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Keywords: Lean tools, Standard work, Line balancing, Visual management, 5S, Production line, Production improvement, Efficiency.

1. Introduction

The automotive industry continues to be an engine of the world economy, presenting consistent growth whenever there are no fears that affect the global economy. Statistics issued by the OICA - International Organization of Motor Vehicles Manufacturers for 2017 [1] show that there was a growth of 2.36% in car production, taking into account the previous year, reaching a global value of 97 million vehicles. While the automotive industry is based on innovation and quality, the automotive components industry is based on three main pillars: competitiveness, delivery

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time and quality [2, 3]. The component industry is still stratified at different levels depending on the direct or indirect way you feed the main production lines [4].

This work was carried out in a company belonging to the automotive components industry, dedicated to the production of high fluid systems, for air conditioning systems in vehicles. The goal of this work was to implement improvements in a production line to increase its productive capacity. The production line was at the beginning of series fabrication, with a production capacity of 1200 parts/day, and an OEE of 53%. To reduce waste and increase the efficiency and productivity of the production line, several continuous improvement techniques, such as line balancing, standard work, visual management and 5S have been implemented. The target of the work was to increase the production capacity to 1800 parts/day, which corresponds to the daily objective of the project, according to PMP (Production Master Plan).

2. Literature review

The competitiveness installed between the different brands in the automotive sector requires systematic procedures for continuous improvement [5, 6]. Eliminating waste, lowering costs and increasing quality, are the day-to-day of these companies [7]. Achieving these improvements depends on many factors, which may be related to the cost of labour and the investment capacity of each company. Continuous improvement in this sector can take two different approaches, depending on the investment that one wants or can make, and the depth of the actions that one intends to implement. Some authors have promoted disruptive situations based just on technical improvements, moving from a purely manual system based on intensive labour, to a completely automated one, based on capital intensive and requiring higher investments, but promoting a substantial reduction in cycle time and a significant improvement in quality [4,8,9]. In other cases, although the improvements mainly focus on the technical improvement, the effects are also felt in the management component of each production cell [10,11]. However, there are approaches that require practically no technical improvements, and very small investments, which, through management tools related to Lean Manufacturing, promote significant process improvements, which eliminate wastes and improve the final quality of the product [12-15]. It can be said that all Lean tools are used extensively in the automotive industry, but VSM (Value Stream Mapping), Line Balancing, Standard Work, SMED, Visual Management and 5S, are the ones that allow results to be obtained faster and with a normally very low level of investment [16-20].

Faced with problems of productivity and repetitive tasks that it was necessary to eliminate, Araújo et al. [4] developed a new concept of manufacturing suspension mats for the automotive industry, from a semi-automatic system to a fully automatic system, achieving 40% productivity gains. This important gain in productivity was due to a restructuring of the layout of production equipment, which allowed to allocate two employees to other more important tasks, eliminating the need for unnecessary intermediate manipulations. The flexibility of the production equipment was also improved. Otherwise, Costa et al. [9] developed a fully automated system for the assembly of systems used to drive the car's windscreen wipers, dramatically increasing flexibility and production control for a family of more than 20 variant parts. Rather than lowering cycle time, the study focused on increasing flexibility, allowing rapid change of the type of workpiece to be assembled, without the setups needing to allocate highly specialized workers over long periods of time. On the other hand, Costa et al. [7] developed a new concept of equipment for the assembly of shafts used in the operation of car windshields. This new concept of assembly allowed to eliminate the appearance of persistent defects related to the manual assembly of these assemblies. In addition to the significant increase in quality and 100% inspection of the assemblies produced, the production rate was also increased compared to the initial semi-automatic equipment by 19%. In order to improve cycle time and concentrate on different assembly activities performed on different equipment, in a single equipment, Moreira et al. [8] developed a new integrated assembly concept. This equipment essentially aimed to eliminate different stages of production, as well as the consequent intermediate stocks. The equipment allows a continuous flow, minimizing the efforts of the internal logistics - supply chain. Nunes et al. [3] developed very flexible equipment capable of assembling a family of sets of parts in an extremely flexible way, achieving a cycle time reduction of more than 35%. The extremely fatiguing manual assembly operations were eliminated, reducing the risk of occupational diseases such as tendonitis. Magalhães et al. [10] eliminated the problems of randomness caused by the operation of cutting wires used in car seats, through an innovative system of separation and alignment, provided with artificial

vision. This made it easier to pass through these components for the next operation - plastic over injection. In order to improve the productive system in tire construction, Santos et al. [11] promoted a technical upgrade of equipment designed for this task, improving cycle time and eliminating systematic downtime due to malfunctions. The safety of the operator has also been strengthened by the new technical solutions. Still considering tire manufacturing, Barbosa et al. [12] studied an extrusion process of a component used in the construction of tires. By applying the DMAIC methodology and a statistical control of production, it was able to reduce manufacturing defects, improving quality indices by about 41%. Through the application of Lean methodologies, Rosa et al. [13–15] significantly improved the production of control cables used for the operation of doors and windows in automobile vehicles. Through different jobs, SMED methodologies were applied, VSM analysis was carried out and the principles of Standard Work were applied. Significant productivity gains were achieved in a product that normally has low added value and uses labor intensive in its assembly. Stadnicka and Litwin [16] added VSM analysis to System Dynamics Analysis (SDA) because it was reported that VSM analysis alone was not enough to eliminate all wastes in a car production line. The authors showed that this integration works effectively. Focusing on the automotive industry, Sternatz [17] reported that it is difficult to meet all external requirements that affect the balancing of a production line. To try to solve this problem, we developed multi-Hoffmann heuristics of Fleszar and Hindi, improving the best existing procedures so far. In order to overcome problems of workload fluctuation in production lines, Fisel et al. [18] developed an algorithm capable of combining different balancing techniques. The model consists of distributing the times and tasks along the line based on workload fluctuations. The model allows the manager to make decisions faster on the line balancing and to avoid the arrival of materials to the line that are not yet necessary, thus saving resources. In order to match the feeding of parts with the balancing of a production line, Nourmohammadi et al. [19] developed a model based on supermarkets to feed the lines, which allowed a reduction of costs related to the storage of components and line balancing. On the other hand, Rosio et al [20] proposed a assessment criterion of existing manufacturing systems, with a view to analyzing their reconfigurability, in order to streamline the response in cases of product change or demand variation. Taking into account the need to improve the productivity of a car air-conditioning systems manufacturing line, Antonioli et al. [6] studied a production line, selected a new pipe folding machine that was integrated in that line, standardized the operations and proceeded to line balancing, increasing the OEE (Overall Equipment Efficiency) by 16%.

As in the work developed by Antonioli et al. [6], the main objective of this study was the application of Lean methodologies to increase the OEE of a car air-conditioning systems production line, initially set at 53%, increasing the production rate from 1200 parts/day to 1800 parts/day.

3. Methodology

The productive flow of this production line is shown in Fig. 1. There are operations for bending the aluminium pipes following the geometry required by the customer, as well as operations for crimping and joining the aluminium pipes with a flexible rubber hose. There are also two quality controls to ensure that the part has no obstruction or leakage. In order to reach the goals above-mentioned, a strategy was drawn, which can be seen in Fig. 2.

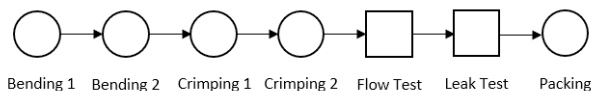


Fig. 1. The productive flow of the production line.

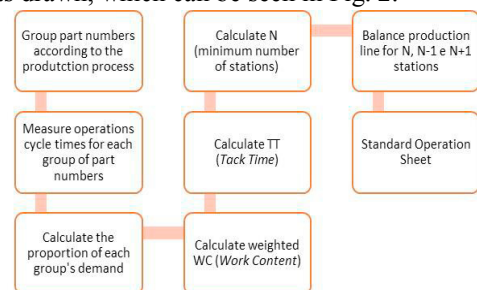


Fig. 2. Approach drawn to reach the desired optimization results.

3.1. Group part numbers according to the production process

In this production line, various part numbers are produced, however, there are many similarities among them in the productive process. The part numbers are grouped according to their production process.

3.2. Measurement of operations cycle time

The definition of groups is followed by the measure of each operation cycle time. Since the part numbers are already grouped, it is unnecessary to measure the cycle time for all part numbers. Therefore, it is necessary to measure the operations cycle time for only one part number per group. The measurement of the cycle time should include the following assumptions:

- The operator must work at a normal pace;
- The operator must be qualified for the position;
- The start and end of the operation cycle must be defined;
- 10 to 20 equal and repetitive cycles must be recorded;
- Number of operator steps must be calculated;
- Waiting times must be recorded;
- Machine times must be recorded.

3.3. The proportion of each group's demand

This proportion represents the productive mix of the production line. Due to the cycle times differences in each group, this ratio allows to calculate the weighted cycle time, which should be less than the takt time of the production line.

3.4. Weighted Work Content

The Work Content (WC) represents the total time required to produce a reference. Since two part numbers are produced simultaneously in the same production line, this value must represent the total time of the operations performed on the line.

3.5. Tack Time

The Takt Time (TT) represents the production rate required to respond to the customer demand. It depends on the demand and availability of the production line. The demand is given by the PMP (Production Master Plan), which dictates the quantities to be produced, maintaining a constant production. The availability of the production line corresponds to the production available time, excluding planned intervals and stops.

3.6. N - Minimum number of stations

The value of N represents the minimum number of stations required for the Cycle Time (CT) of the production line to be lower than the TT.

3.7. Balance production line for N , $N-1$ e $N+1$ station

After the calculation of N , the production line is balanced for N , $N+1$ and $N-1$ stations (each station is represented by one operator). Thus, the production line has the flexibility to function with a different number of operators and level its production rate (when there is an excess or stock deficit). The production capacity depends on the cycle time of the bottleneck station of the production line, which means the station with the lowest cycle time. It

is calculated according to Equation 1, and it is multiplied by 2 since the production line is able to produce two part numbers at the same time, and it is also multiplied by 90% (to consider 10% of losses in production efficiency).

$$\text{Production Capacity} = \frac{3600}{CT} \times 2 \times 90\% \quad (1)$$

3.8. Standard Operation Sheet

The Standard Operation Sheet is the standard document placed on the production line for operators and supervisor consultation, showing the result of the line balancing and contains the following information:

- Operations to be performed at each station;
- Key points and safety points for each operation;
- The standard time for each operation;
- Work content;
- Takt time;
- Cycle time;
- Production target (by hour and shift).

4. Results

4.1. Group part numbers according to the production process

In table 1, there is a matrix with the production process of each part number. The cycle time to assemble the components bracket or soft nose is similar, so the two processes are in the same column.

Table 1. Production process matrix for each part number of the production line.

Part Number	Bending	Crimping	Manual Crimping	Bracket/Soft Nose	Flow Test	Leak Test	Group
A	X	X		X	X	X	G1
B	X	X		X	X	X	G1
C	X	X		X	X	X	G2
D	X	X		X	X	X	G2
E	X	X		X	X	X	G2
F	X	X		X	X	X	G2
G	X	X			X	X	G3
H	X	X			X	X	G3
I	X	X			X	X	G3
J	X	X			X	X	G3
K	X	X	X	X	X	X	G4

4.2. Measurement of operations cycle time

The results of the measurements of the cycle time for each operation is shown in Table 2.

Table 2. Production process matrix for each part number of the production line.

Operation	G1	G2	G3	G4
Bending 1 (s)	26	26	26	26
Bending 2 (s)	42	20	20	-
Crimping 1 (s)	25	25	25	25
Crimping 2 (s)	28	28	28	28
Manual Crimping (s)	-	-	-	14
Soft Nose / Bracket Assembly (s)	14	14	-	14
Flow Test (s)	10	10	10	10
Leak Test (s)	35	45	45	45
Packing (s)	45	35	35	35

4.3. The proportion of each group's demand

In Table 3 is shown the proportion demand for each group.

Table 3. The proportion of each group's demand.

Group	Annual Demand (parts)	Proportion (%)
G1	112781	56%
G2	65226	32%
G3	6369	3%
G4	16502	8%
Total	200878	100%

4.4. Weighted Work Content

Table 4 shows the WC result for each group.

Table 4. Work content for each group and its proportion in the production line.

Group	WC (s)	Proportion (%)
G1	348	56%
G2	348	32%
G3	334	3%
G4	342	8%

The weighted WC is calculated by Equation 2.

$$\begin{aligned} \text{Weighted Work Content} &= \sum WC_i \times \text{Proportion } (\%)_i = \\ &= 348 \times 56\% + 348 \times 32\% + 334 \times 3\% + 342 \times 8\% = 347 \text{ s} \end{aligned} \quad (2)$$

4.5. Tack Time

The weekly availability of each shift is given in Table 5. As the line produces two part numbers in parallel, it is considered the availability of both sides of the production line (high pressure and low pressure), being the weekly availability the sum of both sides of the production line.

Table 5. Weekly availability of the production line.

Shifts	Availability HP (h)	Availability LP (h)	Production Line Availability (h)
1 Shift	38,75	38,75	77,5
2 Shifts	77,5	77,5	155
3 Shifts	111,5	111,5	223

For the calculation of the TT, it was considered the availability of three production shifts and the demand of 1800 parts/day. The TT is calculated according to Equation 3.

$$TT = \frac{\text{Availability}}{\text{Demand}} = \frac{223 \text{ h} \times 3600 \text{ s}}{1800 \text{ parts} \times 5 \text{ days}} = 89 \text{ s} \quad (3)$$

4.6. N - Minimum number of stations

This value was calculated from Equation 4, obtaining the result of 3.9 workstations, which means that at least 4 workstations are needed.

$$N = \frac{\text{Weighted Work Content}}{\text{Tack Time}} = \frac{347 \text{ s}}{89 \text{ s}} = 3,9 = 4 \quad (4)$$

4.7. Balance production line for N, N-1 e N+1 station

As in this case, the result of N=4 was obtained, then it is necessary to balance the production line for 3, 4 and 5 operators. Before the production line balancing, there were four people working in this production line. Station 1 was the bottleneck of this line, with a cycle time of 114 s, which represents a production capacity of 57 parts/hour, as shown in Fig. 3 and Table 6.

To balance the production line, the operations were rearranged to reduce the CT of the bottleneck station. After the line balancing, Station 2 became the bottleneck of the production line, with a CT of 94 s, and a production capacity of 69 parts/hour, as shown in Fig. 4 and Table 7. Although the cycle time was reduced to 94 s, this CT is still above the TT of 89 s, which means the production line capacity is not able to comply with customer demand.



Fig. 3. Cycle time of each station before line balancing.

Table 6. Production capacity (before line balancing).

Cycle Time (s)	114
Production Capacity 90% (parts/h)	57

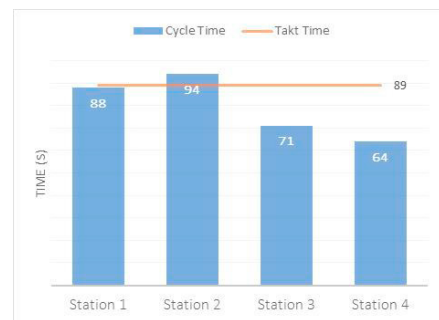


Fig. 4. Cycle time of each station after line balancing with 4 operators.

Table 7. Production capacity (after line balancing with 4 operators).

Cycle Time (s)	94
Production Capacity 90% (parts/h)	69

As mentioned in the methodology, the production line was balanced for N-1 and N+1 workstations, which means three and five workstations. The results for the three workstations are shown in Fig. 5 and Table 8, and the results for five workstations are shown in Fig. 6 and Table 9.

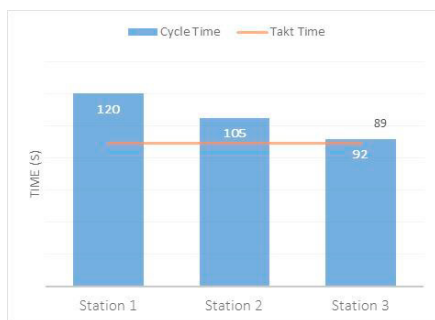


Fig. 5. Cycle time of each station after line balancing with 3 operators.

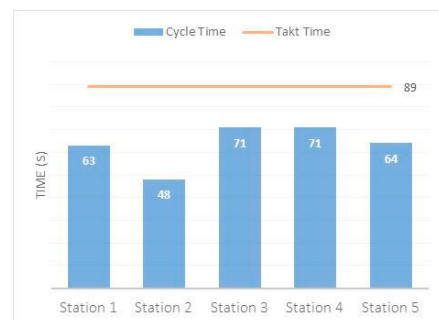


Fig. 6. Cycle time of each station after line balancing with 5 operators.

Table 8. Production capacity (after line balancing with 3 operators).

Cycle Time (s)	120
Production Capacity 90% (parts/h)	60

Table 9. Production capacity (after line balancing with 5 operators).

Cycle Time (s)	71
Production Capacity 90% (parts/h)	91

Fig. 7 shows the production capacity results obtained after the production line balancing with three, four and five operators. Through this methodology, it was possible to considerably increase the production capacity from 57 to 69 parts/h (with four operators), and 91 parts/h (with five operators). The increase in production capacity with the four operators corresponds to an increase of 17% in productivity and an increase of 37% with five operators, as shown in Table 10.



Fig. 7. Production capacity before and after the production line balancing

Table 10. Production capacity before and after line balancing

	Before	After	
Quantity of operators	4 op.	4 op.	5 op.
Production Capacity (parts/h)	57	69	91
Production Increase (%)	-	+17%	+37%

5. Discussion

Regarding Figure 2, it is possible to realize that following the proposed methodology it is possible to obtain really interesting knowledge about the production line and how to balance it, cutting wastes and standardizing tasks. The methodology followed is different from that used in other studies, but it has also obtained interesting results. In a similar work, Rosa et al. [14] achieved an improvement in the production rate by 43%, also in products with relatively low added value, but in this case mainly due to the cutting of waste. On the other hand, a study elaborated on another production line of the same company by Antonioli et al. [6], based mainly on standard work tool, allowed an increase in OEE of 16%, which is in line with the now obtained, of 21%. Another study elaborated by Rosa et al. [13], mainly based on VSM analysis and cutting desertification, also allowed an increase in productivity of 41%. Thus, it is possible to affirm that the methodology followed in this study allows to achieve results within the expected in the balancing of production lines.

6. Conclusions

The objective of this work was to increase the productive capacity of a production line, in order to comply with customer demands. Several lean and continuous improvement methodologies were used to reduce waste and increase productivity. The main tool used was line balancing, by levelling the loads between stations in the production line, and thus reduce the cycle time of the line. Through these methodologies it was possible to increase the production capacity to 2000 parts/day (with one more operator), thus achieving the previously established goal of 1800

parts/day (Table 11). In addition, to increase the production line capacity, this work has led to an increase in line efficiency, waste reduction, and has also increased the production line flexibility by establishing a standard work with three, four and five operators. These results also allowed the increase in the production line OEE to 74%, which results in an improvement of 21%.

Table 11. Comparison between established target and obtained result.

	Target	Result
Production Capacity (parts/day)	1800	2000

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